

EF5342 Financial Systems, Markets and Instruments

Week 6 — Market Efficiency

Prof. Weikai Li

City University of Hong Kong

Semester B 2026–2027

Market Efficiency: The Most Important Debate in Finance

Why does this debate matter more than any other in finance? Because the answer determines whether prices do their job — guiding the allocation of capital — and whether anyone can consistently beat the market. **Agenda**

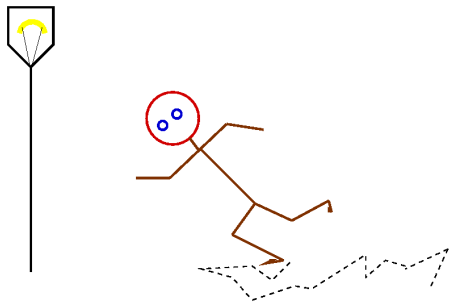
- Efficient Markets Hypothesis
- Evidence for and against market efficiency
- Anomalies: facts and explanations
- Smart Beta ETFs

Efficient Markets Hypothesis (EMH)

Efficient market Prices of securities **fully reflect all available information**

Competition among investors ensures that new information gets incorporated into prices right away. Since new information is unpredictable and random by definition, so are price changes:

- Price changes should follow a **random walk**



Real stock prices look much like a simulated random walk — hard to tell them apart.

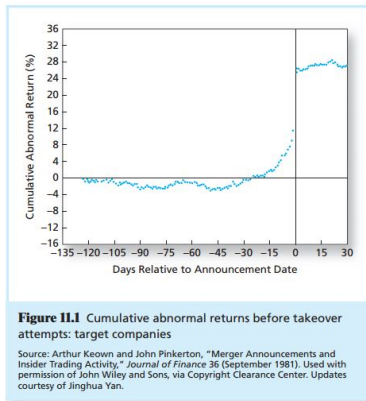
Timely responses imply efficient markets: cumulative abnormal return (CAR) *jumps* on the announcement day for good news (up) and bad news (down), with **no further drift** afterwards. Key feature of efficient markets: *no predictability*.

Untimely responses imply inefficiency:

Leakage Prices react *too soon* — they drift up before the news announcement, as insiders learn the information

Drift Prices react *with a delay* — they keep drifting in the news direction *after* the announcement

Stock Price Reaction to Takeover Announcement



Question: why do prices trend *up* before the takeover announcement?

Case: EntreMed (Huberman & Regev 2001)

- **27 Nov 1997:** *Nature* reports a cancer-research breakthrough, mentioning ENMD as owning licensing rights
- **3 May 1998:** Sunday *New York Times* reports the same story on the front page of the business section
- **12 Nov 1998:** *Wall Street Journal* reports that the results in the 3 May Times article cannot be independently replicated

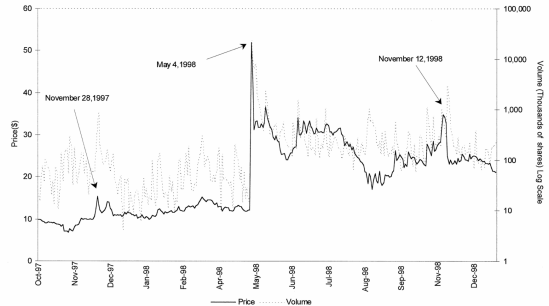


Figure 1. ENMD closing prices and trading volume, October 1, 1997, to December 30, 1998.

Why Do We Care About Market Efficiency?

Deviations from efficiency may offer profitable opportunities to better-informed traders — *at the expense of less-informed traders.*

Inefficient markets $\Rightarrow$ mispriced securities $\Rightarrow$ suboptimal resource allocation:

- **Overvalued** firms can raise capital too cheaply and **overinvest**
- **Undervalued** firms may pass up profitable opportunities because their cost of capital is too high

Thus, **efficient security prices help allocate resources efficiently.**

Weak-Form Market Efficiency

Market efficiency is a matter of **degree** — it depends on how we define “available information.” Three increasing degrees: weak, semi-strong, strong.

Weak form Stock prices already reflect all information contained in the **history of past prices and trading volumes**

Reasoning: if such data ever conveyed reliable signals, investors would have learned to trade on them, impounding them into prices and removing further profit opportunities — a buy signal results in an immediate price increase.

Technical analysis uses historical prices and volume to predict future price changes.
Weak-form efficiency $\Rightarrow$ *technical analysis should not work.*



Semi-Strong Form Market Efficiency

Semi-strong form Prices reflect all **publicly available information**: historical price/volume plus public information on firm fundamentals — revenues and costs, balance-sheet strength, patents held, quality of management, etc.

Reasoning: if useful, investors have learned to trade on this information, already impounding it into prices and removing further profit opportunities.

Fundamental analysis tries to identify mispriced securities by examining economic/financial factors. Semi-strong efficiency $\Rightarrow$ *fundamental analysis should not work*.

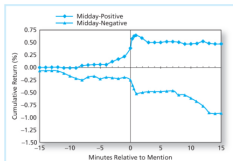


Figure 11.2 Stock Price Reaction to CNBC Reports. The figure shows the reaction of stock prices to on-air stock reports during the "Midday Call" segment on CNBC. The chart plots cumulative returns beginning 15 minutes before the stock report.

Source: Reprinted from J. A. Baise and T. C. Green, "Market Efficiency in Real Time," *Journal of Financial Economics* 63 (2002), p. 422. Copyright 2002 with permission from Elsevier Science.

Strong-Form Market Efficiency

Strong form Prices reflect **all relevant information**: historical and public information *plus insider information* (e.g., upcoming M&A and earnings news)

Reasoning: insiders (or their allies) have already traded on this information, impounding it into prices and removing further profit opportunities. Strong-form efficiency $\Rightarrow$ *trading on insider information is also fruitless.*

Insider Trading — and the Congress

In financial markets, trading on insider information is highly profitable — and insiders may not just be corporate managers.

- Ziobrowski et al. (2004) document abnormal profits earned from mimicking the trades of U.S. senators
- Retail products now track congressional trading: the **Unusual Whales Subversive Democratic Trading ETF** (ticker NANC, Cboe BZX; $\approx$ \$278M AUM, 0.72% expense ratio as of Aug 2026)

Objective

The investment seeks to achieve long-term capital appreciation. The fund is an actively managed diversified exchange-traded fund (“ETF”) that seeks to achieve its investment objective by investing primarily in equity securities of publicly traded companies that sitting Democratic members of United States Congress and/or their families also have reported to have invested in through public disclosure filings made by such Congresspersons pursuant to the Stop Trading on Congressional Knowledge Act (“STOCK Act”).



Three Forms of Market Efficiency: Summary

Form	Information reflected	Implication
Weak form	Historical price/volume	Technical analysis fruitless
Semi-strong	+ all public information	Fundamental analysis fruitless
Strong form	+ insider information	Insider trading fruitless

Each form nests the previous one — the information set only grows.

Evidence Against Market Efficiency

It is not surprising that the market is not strong-form efficient:

- The ability of insiders to trade profitably is well documented by academic studies
- The SEC closely monitors and prosecutes insider trading cases

In a well-functioning market, we should at least expect **semi-strong-form** efficiency. Yet recent studies document evidence that the stock market is not even weak-form efficient: both historical prices/volume *and* current firm fundamentals can predict future stock movements.

Violation of Weak-Form Efficiency: Momentum

Jegadeesh & Titman (1993): stocks with good (bad) past 6- to 12-month returns continue to do well (poorly) in the future.

Why is this a violation of weak-form efficiency? The strategy uses *only* past returns — information already contained in prices — yet earns abnormal profits.

Source: my calculation based on Kenneth French's data.

Violation of Semi-Strong Form Efficiency: Post-Earnings Announcement Drift

Firms announcing unexpectedly **good** (bad) earnings continue to do well (poorly) over the next quarter.

- The market appears to adjust to the earnings surprise only *gradually*, resulting in a sustained period of price drift

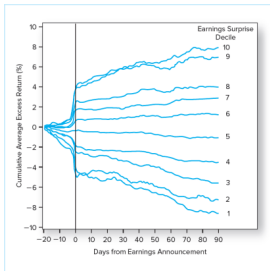


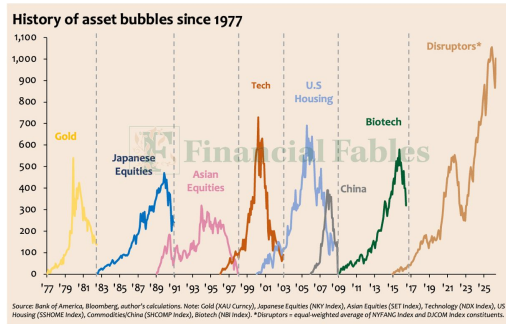
Figure 11.5 Cumulative abnormal returns in response to earnings announcements

Source: Reprinted from R. J. Rendleman Jr., C. P. Jones, and H. A. Latané, "Empirical Anomalies Based on Unexpected Earnings and the Importance of Risk Adjustments," *Journal of Financial Economics* 10 (1982), pp. 269–287.

Asset Bubbles

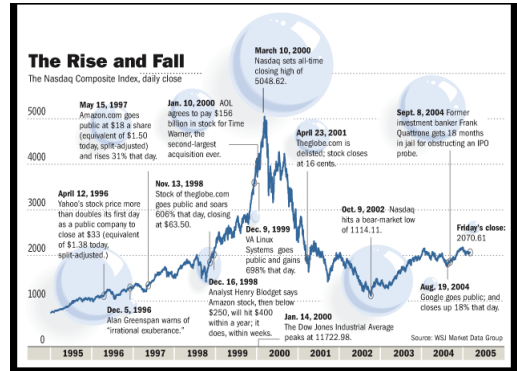
Asset bubbles: for a significant period, an asset trades at a price substantially higher than its fair value — the most dramatic form of market inefficiency and investor irrationality.

Historical episodes, indexed to 1 at start: Tulip Mania (1634–1637, green; peaks around 50–60x); Japanese stocks (1982–1992, light blue; ~80x at peak); Tech/Dot-com (1994–2002, cyan; ~20–30x); Mississippi Bubble (1718–1720, blue; 20–30x); South Sea Bubble (1719–1721, purple).



The Dot-Com Bubble in One Chart

- NASDAQ rose ~400–600% from 1995 to the peak (1,000 → 5,000+)
- Fell ~78% from peak to trough (2000–2002)
- Many dot-com companies went to zero; even survivors like Cisco took ~15 years to recover their 2000 highs



Mini Case: The Dot-Com Bubble (1995–2000) — Irrational Exuberance or Rational Revolution?

Catalyst. Commercialization of the World Wide Web (early 1990s) promised to reshape commerce; firms were valued on “eyeballs” (web traffic), user growth, and platform potential rather than earnings, cash flows, or assets.

EMH viewpoint: the market rationally priced the *option value* of dominance in a winner-take-most digital economy; high uncertainty justified high valuations.

Discussion: in 1998, was it rational to value Amazon (never profitable) above established retailers like Barnes & Noble? Why or why not?

The Bubble Inflation Mechanism (1998–2000)

- **Venture capital fuel:** easy money into startups with “.com” in the name
- **IPO frenzy:** minimal-revenue firms went public with first-day pops of 100%+ (VA Linux: +698% on debut)
- **Retail mania:** online brokerages (E*Trade) and day trading became mainstream
- **Media and analyst hype:** valuation metrics dismissed as “old economy” thinking; “this time is different”
- **Momentum investing:** rising prices attracted buyers, decoupling prices from any plausible fundamentals

The Prick and Collapse (2000–2002)

10 March 2000: NASDAQ peaks at 5,048.62 (up from ~ 750 in 1995). Catalysts: Fed **interest-rate hikes**; **profit warnings** from internet infrastructure firms (e.g., Cisco); **lock-up expirations** letting insiders finally sell; growing **skepticism** (Barron's "Burning Up," March 2000, on cash burn rates). **Aftermath:** NASDAQ fell $\sim 78\%$ by October 2002; most dot-coms failed; survivors (Amazon, eBay) saw $90\%+$ drawdowns; $\sim \$5$ trillion in market value evaporated.

A Failure of Market Efficiency?

Yes: investor irrationality (overconfidence, herd behavior, extrapolation bias); agency problems (analysts at investment banks faced underwriting conflicts and issued overly optimistic reports); asymmetric information (VCs and founders understood the fragility but had incentives to promote the narrative). “Irrational exuberance” — Alan Greenspan (1996), years before the peak.

But consider the counterargument: *“The Dot-Com Bubble proves markets are inefficient because prices clearly deviated from intrinsic value. However, it also proved efficient in the long run by rapidly allocating massive capital to a transformative technology, allowing the optimal business models to emerge through a Darwinian process. Which perspective is more compelling, and what does that tell us about the time horizon of market efficiency?”*

Market Efficiency and Money Making: Mutual Funds

Another test of semi-strong efficiency: the performance of mutual fund managers, most of whose profits would come from exploiting market inefficiency.

- If the market is not semi-strong efficient and managers have skill, they should *consistently* outperform the index benchmark
- Efficient markets: no one can earn abnormal profits *consistently*

Active Funds' Success Rate by Category (%)

Category	1-Year	3-Year	5-Year	10-Year	15-Year	20-Year	10-Year (Lowest Cost)*	10-Year (Highest Cost)
U.S. Large Blend	41.4	33.5	20.6	9.5	9.4	9.3	19.2	1.3
U.S. Large Value	34.1	37.0	21.9	14.0	12.0	13.4	27.5	9.8
U.S. Large Growth	31.9	31.5	31.8	8.2	6.2	5.5	18.0	2.4
U.S. Mid Blend	66.7	30.0	18.8	16.9	7.7	6.6	29.2	4.0
U.S. Mid Value	26.8	53.2	37.1	10.7	25.6	—	9.1	29.8
U.S. Mid Growth	46.0	52.2	58.1	37.9	22.3	—	41.5	32.6
U.S. Small Blend	58.6	38.1	23.2	17.2	12.3	19.8	31.4	13.9
U.S. Small Value	31.3	41.4	38.2	20.2	26.9	17.9	20.0	5.8
U.S. Small Growth	47.4	65.8	55.1	44.0	27.9	12.9	48.8	48.8
Foreign Large Blend	43.7	51.4	38.5	31.6	29.1	23.2	40.6	13.4
Foreign Large Value	33.7	54.7	35.6	29.8	11.4	—	26.9	14.3
Foreign Small-Mid Blend	48.1	48.5	31.3	31.8	—	—	25.0	41.8
World Large Blend	38.9	26.5	18.9	16.7	—	—	22.2	10.8
Diversified Emerging Markets	30.1	61.3	47.4	54.4	40.7	—	68.0	53.3
Europe Stock	53.3	73.7	50.0	42.9	37.9	21.8	80.0	16.7
U.S. Real Estate	82.5	83.6	64.2	48.5	27.0	41.1	50.0	33.8
Global Real Estate	83.3	72.2	62.9	50.0	—	—	33.3	50.0
Intermediate Core Bond	69.2	47.0	43.2	37.8	17.0	8.8	54.8	20.6
Corporate Bond	64.2	47.5	26.0	28.9	—	—	33.3	25.8
High-Yield Bond	72.0	56.1	44.7	55.3	—	—	26.8	39.7

Sources: Morningstar. Data and calculations as of Dec. 31, 2021. *Green shading indicates that active funds in this fee quintile had above/below-average success rates.

Outperforming becomes harder the longer the evaluation window. Success rate: fraction of active funds across the 20 Morningstar categories that survived and outperformed their average passive peer.

Patterns in stock prices that cannot be easily explained by the EMH are called **anomalies**. Practitioners are very interested — anomalies are potential money-making machines. Recent decades have produced hundreds of documented anomalies. The most robust:

- Size effect
- Value premium
- Long-term reversal
- Low volatility anomaly
- Profitability / quality
- Investment
- Share issuance

Sort stocks into 10 portfolios by **market capitalization**: small firms (decile 1) earn higher average returns than large firms (decile 10).

Source: my calculation based on Ken French's data.

Value Premium — and Competing Explanations

Sort all stocks into 10 portfolios by **book-to-market**: value stocks (decile 10) outperform growth stocks (decile 1) on average. (Source: my calculation based on Ken French's data.)

Why does the premium exist?

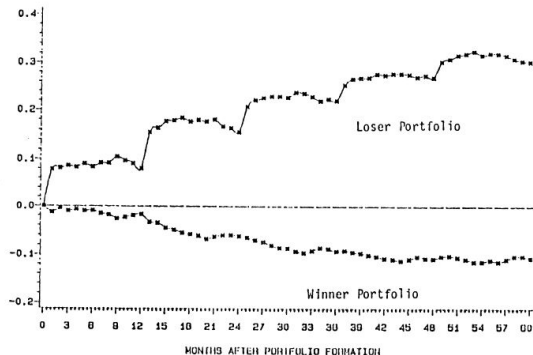
Risk-based view Value firms have higher leverage, more irreversible capital, and greater exposure to financial-distress risk in downturns — they are inherently riskier

Behavioral view Lakonishok, Shleifer, and Vishny (1994): investors extrapolate past growth too far, overpricing glamour stocks and underpricing value stocks

Both explanations find some support in the data.

Long-Term Reversal

Stock prices exhibit **long-term reversal**. De Bondt and Thaler (1985) — one of the first anomaly papers — show that stocks that performed poorly over the last 3 years tend to do very well in the following 5 years.



Source: De Bondt and Thaler (1985)

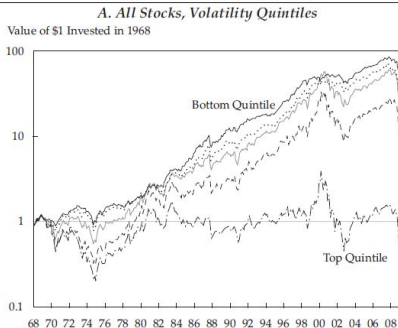
Explanation: investors *overreact* to past firm performance — overly pessimistic about firms with poor past performance, leading to undervaluation. Eventually they realize these loser stocks are not so bad, and prices revert in the long run.

Low Volatility Anomaly

The first principle of finance (and life) is that high risk comes with high return — yet this principle is *often violated in the market* (Baker, Bradley, and Wurgler 2011, *FAJ*).

Explanation: investors' preference for lottery-type stocks

- Investors accept bets with a small chance of winning big but a lower average return: people prefer a 1/1000 chance of winning \$5,000 over \$10 with certainty (Tversky and Kahneman, 1992)
- High-volatility stocks resemble a lottery — and are thus *overvalued*
- Evidence: Kumar (2009) — retail investors who spend more on lotteries also buy more high-volatility stocks



Source: my calculation based on Kenneth French's data.

Profitability, Investment, and Share Issuance

Profitability / quality More profitable firms (quintile 5) earn *higher* returns than less profitable firms (quintile 1). Profitability is part of a broad set of quality characteristics (Asness, Frazzini, Pedersen 2018: profitability, profit growth, low earnings volatility, low credit risk, high dividend yield) — despite being *less* risky, quality stocks earn higher returns

Investment Firms that invest more (quintile 5) earn *lower* average returns than firms that invest less — markets may be too optimistic about high-investment firms, pushing up prices and lowering returns

Share issuance Issuers (top 20%) underperform repurchasers (issuance < 0) — consistent with *market timing*: managers issue when they believe their stock is overvalued, repurchase when undervalued

Source (all three): my calculation based on Kenneth French's data.

Interpreting Anomalies

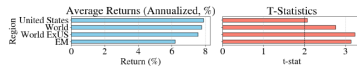
Is a higher return sufficient evidence of an anomaly? **No** — we need positive *alpha* (risk-adjusted return), which raises the question of the correct asset pricing model.

1. **Risk** Anomalies are compensation for taking risks (value and small stocks are riskier) — but hard to justify for profitability and low volatility
2. **Inefficient markets** Investor biases in trading lead to market inefficiency
3. **Data mining** Search data repeatedly and you find patterns by chance; some anomalies cannot be replicated (Hou, Xue, and Zhang, 2018). Real anomalies should exist across different markets/assets — value, size, and momentum may be real

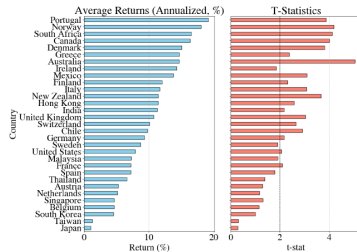


Momentum Is Pervasive: International Evidence

The momentum effect exists in the majority of countries — especially strong in Europe, weaker among Asian countries.



(a) Regional price Momentum.



(b) Country-level price Momentum.

Baltussen, Dom, Van Vliet, and Vidojevic (2025)

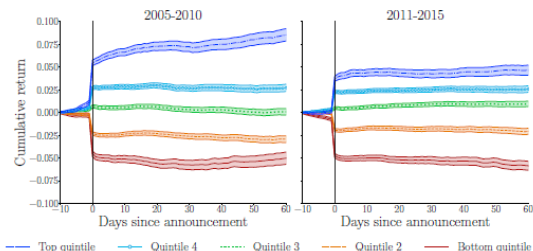
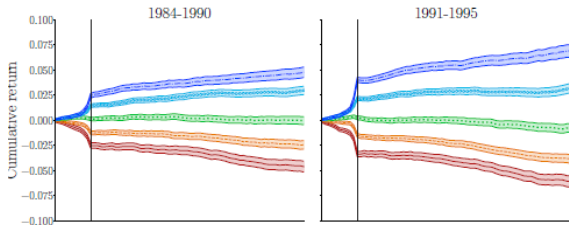
Anomalies May Be Real, But...

...costly to exploit due to trading costs

- Most anomalies exist only in small-cap stocks with high transaction costs and short-selling constraints

...profits may decay over time

- Improved trading technology makes markets more efficient; investors learn; more competitors trade the anomalies



Post-earnings announcement drift across periods — Source: Martineau (2021)

The Paradox of Efficiency

Grossman–Stiglitz paradox If everyone believes markets are efficient, no one will bother to seek information — making prices inefficient. *The very act of trying to beat the market is what makes it efficient.*

The EMH as a lens:

- It is the default null hypothesis; the burden of proof is on the active manager
- Remarkably robust, but not a perfect description of reality
- Finance has evolved from “are markets efficient?” to *how* efficient, under what conditions, and where the persistent inefficiencies are

All these anomalies sound great — but can investors benefit from them?

- In the past: no easy way — only high-net-worth investors could access hedge funds investing systematically in these strategies
- Recently: many **Smart Beta ETFs** allow retail investors to invest in such strategies easily

Smart Beta ETF An exchange-traded fund tracking a *non-market-cap-weighted* index, constructed on transparent, rules-based criteria that tilt portfolio exposures toward one or more investment factors — proven hedge-fund strategies at much lower cost and with great transparency

Smart Beta ETFs: Passive or Active?

Smart beta *blurs the line*: **passive** because it is rules-based, transparent, and low cost relative to active funds; **active** because it makes deliberate choices about index design — which factors to target, how to measure them, how to weight securities, how often to rebalance.

Strategy	Approach	Avg. expense ratio	Factor transparency
Market-cap ETF	Full passive replication	~0.16%	None
Smart Beta ETF	Rules-based factor tilt	~0.33%	High
Active Quant Fund	Model-driven, PM discretion	~0.60%	Moderate
Traditional Active	Stock-picker judgment	~0.62%	Low

Asset-weighted averages, Morningstar 2025 Fund Fee Study (2024 data); smart-beta figure is an issuer-reported average. Cheapest S&P 500 ETFs: SPLG 0.02%, VOO/IVV 0.03%.

The Academic Genesis — Factor Investing

- **CAPM**: only one systematic risk factor — market beta — prices securities in equilibrium
- **Fama and French (1993)**: two additional factors — size (SMB: small minus big) and value (HML: high minus low book-to-market) — explain a large portion of the cross-section of equity returns that CAPM cannot
- **Carhart (1997)**: added momentum (MOM: past-12-month winners minus losers, skipping one month) — the canonical four-factor model
- **Fama and French (2015)**: extended with two factors related to investment and profitability



The Six Core Investable Factors

Despite the proliferation of proposed factors, practitioner consensus — reflected in products from BlackRock iShares, MSCI, and S&P DJI — has converged on a manageable set:

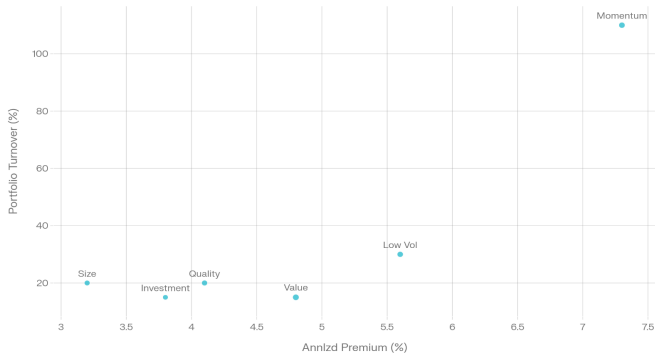
Factor	Definition	Measurement	Economic rationale
Value	Cheap vs. expensive stocks	P/B, P/E, P/S ratios	Risk compensation or mean-reversion of mispricing
Size	Small-cap vs. large-cap	Market capitalization	Liquidity/information risk premium
Momentum	Past winners vs. losers	12–1 month returns	Underreaction, behavioral extrapolation
Quality	High profitability, stable earnings	ROE, accruals, leverage	Distress risk; earnings persistence
Low Volatility	Low-risk stocks	Historical return variance	Lottery demand, leverage constraints
Investment	Conservative vs. aggressive capex	Asset growth, capex	Asset pricing / q-theory

The Six Core Investable Factors

Factor Premia vs Implementation Cost (Turnover)

Source: Research Affiliates / Fama-French | Bubble size = academic confidence

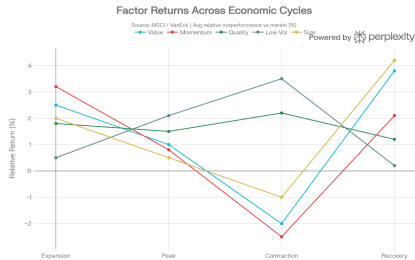
Powered by  perplexity



Factor Cyclicity

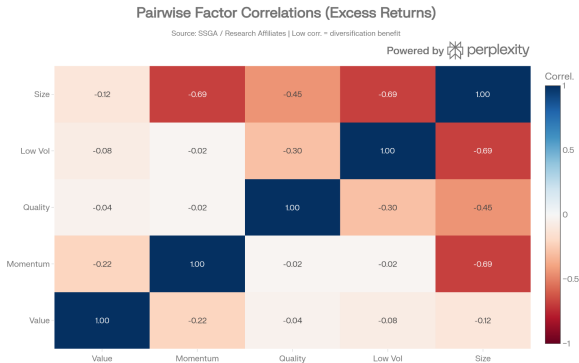
Factor premia are *not constant* — they vary systematically with the macroeconomic and credit cycle:

- **Expansions** (rising growth): Value and Momentum perform best; Low Volatility lags
- **Contractions** (falling growth): Quality and Low Volatility provide defensive outperformance; Value and Momentum suffer sharply
- **Recoveries**: Size (equal weight) and Value tend to outperform as risk appetite returns

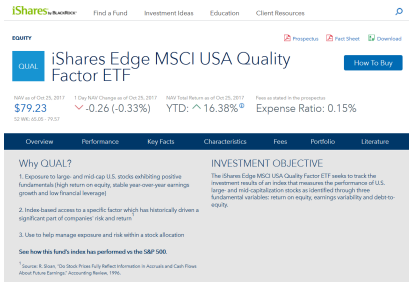


Multi-Factor Diversification

Factors exhibit **low to negative correlations** with each other in excess-return space. Combining them substantially reduces tracking error without proportionately reducing expected returns.



A “Quality” Smart Beta ETF



Growth of Hypothetical \$10,000 ⓘ



A “Momentum” Smart Beta ETF

MTUM

iShares MSCI USA Momentum Factor ETF

Fact Sheet as of 12/31/2022

The iShares MSCI USA Momentum Factor ETF seeks to track the investment results of an index composed of U.S. large- and mid-capitalization stocks exhibiting relatively higher price momentum.

WHY MTUM?

- 1 Exposure to large- and mid-cap U.S. stocks exhibiting relatively higher price momentum
- 2 Index-based access to a specific factor which has historically driven a significant part of companies' risk and return¹
- 3 Use to help manage exposure and risk within a stock allocation

GROWTH OF 10,000 USD SINCE INCEPTION



The Growth of \$10,000 chart reflects a hypothetical \$10,000 investment and assumes reinvestment of dividends and capital gains. Fund expenses, including management fees and

iShares[®]
by BLACKROCK[®]

KEY FACTS

Fund Launch Date	04/16/2013
Expense Ratio	0.15%
Benchmark	MSCI USA Momentum SR Variant Index
30 Day SEC Yield	1.86%
Number of Holdings	125
Net Assets	\$12,047,115,642

Ticker	MTUM
CUSIP	46432F396
Exchange	Cboe BZX formerly known as BATS

TOP HOLDINGS (%)

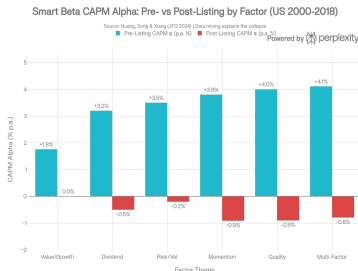
ELI LILLY	5.16
UNITEDHEALTH GROUP INC	5.10
EXXON MOBIL CORP	5.02
CHEVRON CORP	5.00
MERCK & CO INC	3.87
ABBVIE INC	3.39
CONOCOPHILLIPS	2.79
PEPSICO INC	2.78
AMGEN INC	2.35
MCDONALDS CORP	2.28
	37.74

Holdings are subject to change.

Performance Evidence — Theory vs. Reality

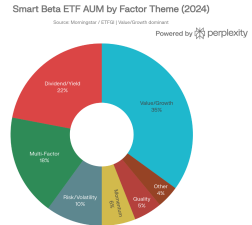
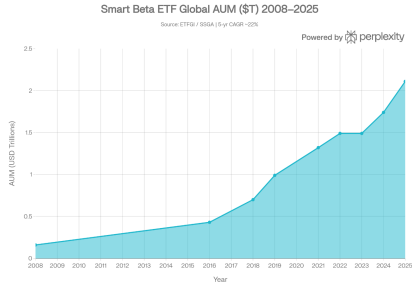
Factor returns look strong in academic backtests but shrink markedly once strategies become investable:

- **McLean and Pontiff (*Journal of Finance*, 2016):** across 97 published anomalies, returns decline by $\sim 26\%$ out-of-sample and by $\sim 32\%$ after publication
- **Hou, Xue, and Zhang (*AER*, 2020):** under replicated tests, only 22% of 452 studied anomalies remain significant



Industry Trends

Smart beta AUM grew from \$160 billion in 2008 to approximately \$2.6 trillion by early 2026 (Morningstar) — a more than sixteenfold increase since 2008.



Portfolio Construction Applications

Core–satellite framework — the most common deployment:

- Passive market-cap *core* (60–80% of equity allocation): low-cost broad-market exposure
- Smart beta *satellites* (20–40%): factor tilts toward desired outcomes — higher expected return, lower volatility, or income

Strategic factor allocation:

- Long-horizon investors (pensions, endowments): value, size, profitability — longest academic documentation, strongest risk-based rationale
- Volatility-sensitive investors: min-volatility and quality reduce downside capture
- Income-focused investors: high-dividend-yield ETFs (value tilt plus current income)

Factor timing — mostly inadvisable: the empirical literature on factor timing is discouraging; rotating on relative factor valuations has not produced consistently strong out-of-sample results. Consensus: maintain diversified, rebalanced multi-factor exposures.

Key Takeaways

EMH Security prices fully reflect all available information; new information is random
⇒ prices follow a random walk and are unpredictable

Three forms Weak (past prices/volume — technical analysis useless); semi-strong (all public info — fundamental analysis useless); strong (all info — no one beats the market, including insiders)

Evidence against Momentum (violates weak form); post-earnings drift (violates semi-strong); asset bubbles (dot-com); size, value, low volatility, profitability, and investment effects

Interpreting anomalies Risk compensation; behavioral biases (overreaction, lottery preference); data mining (non-replicable patterns)

Practical implications Smart beta ETFs let retail investors systematically exploit anomalies via rules-based, transparent strategies; the Grossman–Stiglitz paradox — trying to beat the market is what makes it efficient